

Macroeconomics

Financial Markets and Expectations

(Blanchard, Macroeconomics 8th edition Ch. 14)

Recap

- ▶ So far, we have analyzed various economic outcomes (e.g., output, labor markets, inflation, investment, capital, growth rates) in the short, medium, and long run
 - ▶ Short run: Macroeconomic fluctuations or business cycles assuming that prices are fixed
 - ▶ Medium run: The response of economies where prices can adjust, output may return to potential output
 - ▶ Long run: Economic growth, comparisons across time and between countries; the role of technological progress, savings rate, institutions, etc.
- ▶ Expectations play an important role in the economy: e.g., inflation, for investment (residential or non-residential), exchange rates, interest rates, etc.

Financial Markets and Expectations

- ▶ Our focus will now be on the role expectations play in the determination of asset prices, from bonds, to stocks, to houses.
- ▶ Previously, we discussed the role of expectations informally at various points.
- ▶ It is now time to do it more formally.
- ▶ The **expected present discounted value** of a sequence of future payments is the value today of this expected sequence of payments.
- ▶ Expected present discounted values are not directly observable but must be constructed from information on the sequence of expected payments and expected interest rates.

Expected Present Discounted Values

- ▶ $1/(1 + i_t)$ is the **discount factor** with the **discount rate** i_t , which is used to compute the present discounted value of one dollar to be received next year.
- ▶ The higher the nominal interest rate, the lower the value today if a dollar received next year.
- ▶ Example: The value today of a dollar received two years from now.

Finite Payments

- ▶ The *expected present discounted value* or *present value* of a payment $\$z$ and its expected payment $\$z^e$ in the future:

$$\$V_t = \$z_t + \frac{1}{(1+i_t)} \$z_{t+1}^e + \frac{1}{(1+i_t)(1+i_{t+1}^e)} \$z_{t+2}^e + \dots$$

- ▶ For constant interest rates i :

$$\$V_t = \$z_t + \frac{1}{(1+i)} \$z_{t+1}^e + \frac{1}{(1+i)^2} \$z_{t+2}^e + \dots$$

- ▶ For constant interest rates and constant payments $\$z$:

$$\$V_t = \$z \left[1 + \frac{1}{(1+i)} + \frac{1}{(1+i)^2} + \dots + \frac{1}{(1+i)^{n-1}} \right]$$

- ▶ e.g., fixed rate 30-year mortgage, $n = 30$

Finite Payments (con.)

- From before:

$$\$V_t = \$z \left[1 + \frac{1}{(1+i)} + \frac{1}{(1+i)^2} + \dots + \frac{1}{(1+i)^{n-1}} \right]$$

- This is a geometric series equal to

$$\$V_t = \$z \frac{1 - [1/(1+i)^n]}{1 - [1/(1+i)]}$$

Infinite Payments

- ▶ For constant interest rates and payment forever (starting next period):

$$\$V_t = \frac{\$z}{i}$$

- ▶ If $i = 0$, then the present discounted value of a sequence of expected payments is just the sum of those expected payments.
- ▶ The present value of a sequence of *real* (as opposed to nominal) payments (discounted by real interest rates r_t):

$$V_t = z_t + \frac{1}{(1 + r_t)} z_{t+1}^e + \frac{1}{(1 + r_t)(1 + r_{t+1}^e)} z_{t+2}^e + \dots$$

- ▶ Alternatively, one can deflate:

$$\$V_t / P_t = V_t$$

Expected Present Discounted Values

- ▶ Two ways to compute the present value of payments:
 1. The present value of the sequence of payments expressed in dollars, discounted using nominal interest rates, and then divided by the price level today.
 2. The present value of the sequence of payments expressed in real terms, discounted using real interest rates.

Bonds

► Bonds differ in two basic dimensions:

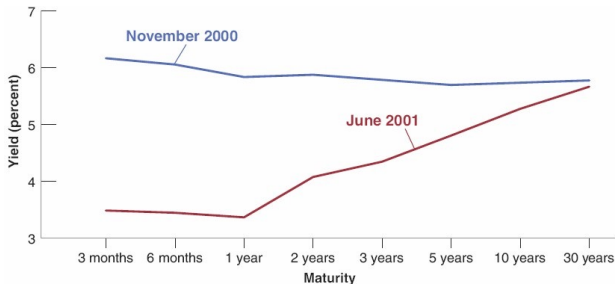
1. Maturity: The length of time over which the bond promises to make payments to the holder of the bond.
2. Risk:
 - 2.1 Default risk as the risk that the issuer of the bond will not pay back the full amount promised by the bond; or
 - 2.2 Price risk as the uncertainty about the price you can sell the bond for if you want to sell it in the future before maturity.

Yields and Interest Rates

- ▶ **Yield to maturity** or **yield**: The interest rates associated with bonds of different maturities
- ▶ **Short-term interest rates**: Yields on bonds with a short maturity, typically a year or less
- ▶ **Long-term interest rates**: Yields on bonds with a longer maturity than a year
- ▶ **Term structure of interest rates** or **yield curve**: The relation between maturity and yield
- ▶ <https://www.youtube.com/watch?v=DCQwiF0J7hw>

US Yield Curves

- ▶ The yield curve, which was slightly downward sloping in November 2000, was sharply upward sloping seven months later.



Source: FRED. Series DGS1MO, DGS3MO, DGS6MO, DGS1, DGS2, DGS3, DGS5, DGS7, DGS10, DGS20, DGS30.

- ▶ Euro area: https://www.ecb.europa.eu/stats/financial_markets_and_interest_rates/euro_area_yield_curves/html/index.en.html

The Vocabulary of Bond Markets

- ▶ **Government bonds:** Bonds issued by the governments
- ▶ **Corporate bonds:** Bonds issued by firms
- ▶ **Bond ratings:** ratings for default risk
- ▶ **Risk premium:** The difference between the interest rate paid on a given bond and the interest rate on the bond with the best rating
- ▶ **Junk bonds:** Bonds with high default risk
- ▶ **Discount bonds:** Bonds that promise a single payment at maturity called the face value

The Vocabulary of Bond Markets (con.)

- ▶ **Coupon bonds:** Bonds that promise multiple payments before maturity and one payment at maturity
- ▶ **Coupon payments:** The payments before maturity
- ▶ **Coupon rate:** The ratio of the coupon payments to the face value
- ▶ **Current yield:** The ratio of the coupon payment to the price of the bond
- ▶ **Life:** The amount of time left until the bond matures
- ▶ **Treasury bills (T-bills):** U.S. government bonds with a maturity up to a year

The Vocabulary of Bond Markets (con.)

- ▶ **Treasury notes:** U.S. government bonds with a maturity of 1 to 10 years
- ▶ **Treasury bonds:** U.S. government bonds with a maturity of 10 or more years
- ▶ **Term premium:** The premium associated with longer maturities
- ▶ **Indexed bonds:** Bonds that promise payments adjusted for inflation
- ▶ **Treasury Inflation Protected Securities (TIPS):** Indexed bonds introduced in the United States in 1997

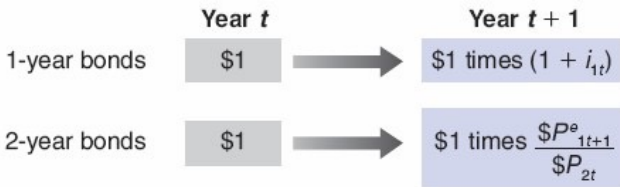
Bond Prices and Bond Yields

- ▶ The price of a one-year bond that promises to pay \$100 next year:

$$\$P_{1t} = \frac{\$100}{1 + i_{1t}}$$

- ▶ The price of a two-year bond that promises to pay \$100 in 2 years:

$$\$P_{2t} = \frac{\$100}{(1 + i_{1t})(1 + i_{1t+1}^e)}$$



Bond Prices and Bond Yields (con.)

- ▶ **Arbitrage:** The expected returns on two assets must be equal.
- ▶ **Expectations hypothesis:** Investors care only about the expected returns and do not care about risk.
- ▶ The two bonds above must offer the same expected one-year return:

$$1 + i_{1t} = \frac{\$P_{1t+1}^e}{\$P_{2t}}$$
$$\$P_{2t} = \frac{\$P_{1t+1}^e}{1 + i_{1t}}$$

- ▶ which means that the price of a two-year bond today is the present value of the expected price of the bond next year.

Bond Prices and Bond Yields (con.)

- ▶ The expected price of one-year bonds next year with a payment of \$100:

$$P_{1t+1}^e = \frac{\$100}{1 + i_{1t+1}^e}$$

- ▶ so that

$$P_{2t} = \frac{\$100}{(1 + i_{1t})(1 + i_{1t+1}^e)}$$

- ▶ which is the same equation as we had before.

Yield to Maturity

- ▶ The *yield to maturity* on an n -year bond (n -year interest rate) is the constant annual interest rate that makes the bond price today equal to the present value of future payments on the bond.
- ▶ The yield to maturity on a two-year bond that satisfies:

$$P_{2t} = \frac{\$100}{(1 + i_{2t})^2}$$

- ▶ is

$$i_{2t} \approx \frac{1}{2}(i_{1t} + i_{1t+1}^e)$$

- ▶ which means that the two-year interest rate is (approximately) the average of the current one-year interest rate and next year's expected one-year interest rate.
- ▶ Long-term interest rates reflect current and future expected short-term interest rates.

Risk Premium

- ▶ Demand risk premium x to hold two-year bond, no arbitrage implies:

$$1 + i_{1t} + x = \frac{\$P_{1t+1}^e}{\$P_{2t}}$$
$$\Rightarrow \$P_{2t} = \frac{\$100}{(1 + i_{1t} + x)(1 + i_{1t+1}^e)}$$

- ▶ so that the two-year yield is:

$$i_{2t} \approx \frac{1}{2}(i_{1t} + i_{1t+1}^e + x)$$

- ▶ which is the average of the current and expected one-year rate plus a risk premium.

Interpreting the Yield Curve

- ▶ If investors expected interest rates to be constant, the yield curve should be slightly upward sloping (risk premium).
- ▶ If they expect interest rates to go down over time, may end up with inverted yield curve (decline in interest rates more than compensates for rising term premium).
- ▶ If they expect interest rates to go up, yield curve should also slope upward.

FINANCIAL STATEMENT: AU LOST ALMOST DKK 300 MILLION LAST YEAR

Aarhus University ended 2022 with a deficit of DKK 292 million according to the annual report recently approved by the board. Losses in the financial markets have been the cause of the significant deficit. The university needs to restore its equity. However, faculties currently in the process of reducing costs will not have to cover the three-digit loss in AU's investments, says the university director.

AU and Bonds (con.)

- ▶ <https://omnibus.au.dk/en/archive/show/artikel/au-tabte-naesten-300-millioner-kroner-sidste-aar>
- ▶ “The financial items have taken a beating. However, we haven't lost more than anyone else if we compare our results with the other universities, and we're at the same level as the pension funds.”
- ▶ “What made 2022 special was that the value of bonds and shares fell simultaneously. This hasn't happened in decades. It seems reasonable to ask how we as a university can lose money on financial items. In this context, it's important to know that we're obliged to invest our surplus liquidity.”
- ▶ “We have to acknowledge that the usual diversification of risks has not been an efficient bulwark in 2022. The bonds were not the safe haven they used to be. When interest rates increase, bond prices fall. So everyone on the market took serious hits on what's normally considered a rather conservative form of investment.”

Inflation-linked Bonds

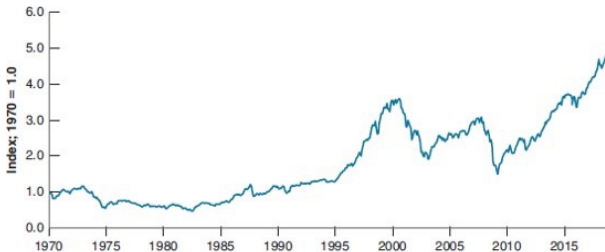
- ▶ <https://www.nationalbanken.dk/en/government-debt/funding-strategy/inflation-linked-bonds>
- ▶ “Over the life of an inflation-linked bond, its principal will vary with consumer prices. At maturity, the redemption amount is the inflation-indexed principal. However, Danish government inflation-linked bonds have a deflation floor, ensuring that investors always, as a minimum, get the full nominal principal back.”
- ▶ “Expressed in per cent, the coupon rate on an inflation-linked bond remains fixed, but as the actual coupon payable is calculated on the basis of the inflation-adjusted principal, the amount will vary with consumer prices as well.”
- ▶ “Danish inflation-linked bonds are held by a broad group of investors, the largest groups being domestic insurance and pensions funds and investment funds.”

The Stock Market and Movements in Stock Prices

- ▶ Firms finance themselves through:
 - ▶ Internal finance: Using their own earnings
 - ▶ External finance: Bank loans
 - ▶ Debt finance: Bonds and loans
 - ▶ Equity finance: Issuing stocks or shares that pay dividends

The Stock Market and Movements in Stock Prices

- ▶ Standard and Poor's stock price index in real terms since 1970:



Source: FRED SP500, CPIAUCSL.

- ▶ Note the sharp fluctuations in stock prices since the mid-1990s.

Movements in Stock Prices

- ▶ Returns from holding one-year bonds or stocks for one year:



- ▶ $\$Q$ is the price of the stock
- ▶ $\$D^e$ is the expected dividend
- ▶ Ex-dividend price: The stock price after the dividend has been paid this year

Equilibrium

- Equilibrium requires that the expected rate of return from holding stocks for one year (left side) be the same as the rate of return on one-year bonds plus the *equity premium* x :

$$\frac{\$D_{t+1}^e + \$Q_{t+1}^e}{\$Q_t} = 1 + i_{1t} + x$$

- or

$$\$Q_t = \frac{\$D_{t+1}^e}{(1 + i_{1t} + x)} + \frac{\$Q_{t+1}^e}{(1 + i_{1t} + x)}$$

Stock Price

- If the expected prices in n years equal the present values of the expected prices and dividends:

$$\begin{aligned} \$Q_t = & \frac{\$D_{t+1}^e}{(1 + i_{1t} + x)} + \frac{\$D_{t+2}^e}{(1 + i_{1t} + x)(1 + i_{1t+1}^e + x)} + \dots \\ & + \frac{\$D_{t+n}^e}{(1 + i_{1t} + x) \dots (1 + i_{1t+n-1}^e + x)} \\ & + \frac{\$Q_{t+n}^e}{(1 + i_{1t} + x) \dots (1 + i_{1t+n-1}^e + x)} \end{aligned}$$

- If the interest rate is positive, then it reduces to:

$$\begin{aligned} \$Q_t = & \frac{\$D_{t+1}^e}{(1 + i_{1t} + x)} + \frac{\$D_{t+2}^e}{(1 + i_{1t} + x)(1 + i_{1t+1}^e + x)} + \dots \\ & + \frac{\$D_{t+n}^e}{(1 + i_{1t} + x) \dots (1 + i_{1t+n-1}^e + x)} \end{aligned}$$

Stock Price (con.)

- ▶ Replacing the nominal interest rates with the real interest rates, then the real stock price is:

$$Q_t = \frac{D_{t+1}^e}{(1 + r_{1t} + x)} + \frac{D_{t+2}^e}{(1 + r_{1t} + x)(1 + r_{1t+1}^e + x)} + \dots$$

- ▶ Implications:
 - ▶ Higher expected future real dividends lead to a higher real stock price.
 - ▶ Higher current and expected future one-year real interest rates lead to a lower real stock price.
 - ▶ A higher equity premium leads to a lower stock price.

Expansionary Monetary Policy

Increase in Consumption Spending

Risk, Bubbles, Fads, and Asset Prices

- Fundamental value: The present value of expected dividends given by

$$Q_t = \frac{D_{t+1}^e}{(1 + r_{1t} + x)} + \frac{D_{t+2}^e}{(1 + r_{1t} + x)(1 + r_{1t+1}^e + x)} + \dots$$

Stocks are sometimes underpriced or overpriced.

- Rational speculative bubbles: Stock prices increase just because investors expect them to.
- Fads: Stocks become high priced for no reason other than its price has increased in the past.

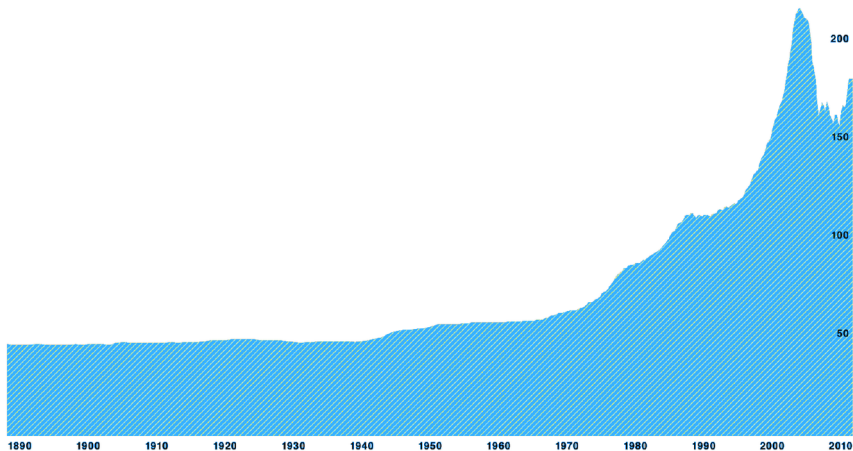
Famous Bubbles

- ▶ In 1634, a “tulip bubble” began to take place as the price of rare bulbs started increasing, and speculators bought tulip bulbs in anticipation of even higher prices later.
- ▶ The price of a bulb called “Admiral Van de Eyck” jumped from 1,500 guineas in 1634 to 7,500 guineas in 1637.
- ▶ In 1994, a Russian “financier” created a company called MMM and promised shareholders a rate of return of at least 3,000% a year.
- ▶ Even though MMM was not involved in any type of production and held no assets, its share prices increased from 1,600 rubles in February to 105,000 in July.
- ▶ Ponzi schemes and dogecoin?

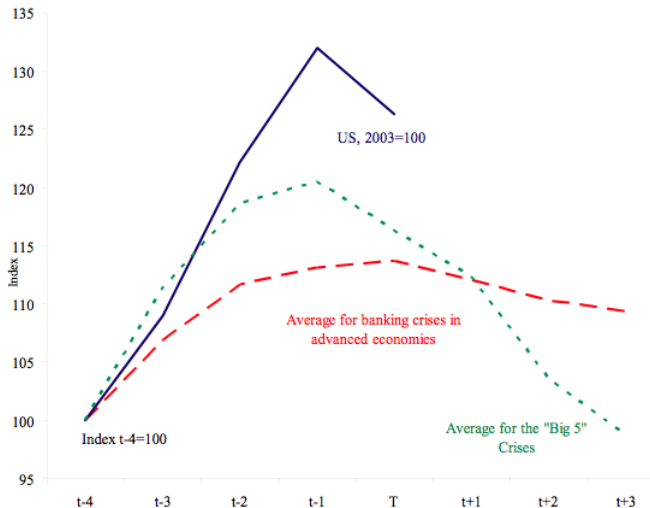
Housing Bubbles

- ▶ Housing prices essentially flat in many developed countries over the last 100 years
 - ▶ Returns mainly from housing services
 - ▶ In general, cannot expect real house price to appreciate much
- ▶ Reinhart and Rogoff (2008): big five crises since World War II in developed countries
 - ▶ Spain (1977), Norway (1987), Finland (1991), Sweden (1991), and Japan (1992), plus 13 other bank-centered crises in developed countries
- ▶ Housing price appreciations very significant in lead up to crises
- ▶ “Big 5”: 20% over four years

Housing Boom: Nominal Home Prices



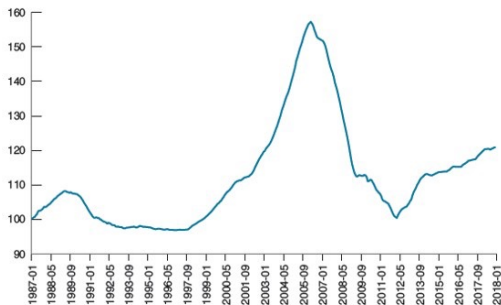
Real Housing Prices and Crises



US Housing Prices

- ▶ Did the rise reflect fundamentals or was it a bubble?
- ▶ Pessimists: house prices should depend on current and expected future interest rates, current and expected rents, and a risk premium.
- ▶ Also, surveys of house buyers suggested extremely high expectations of continuing large increases in housing prices ($> 10\%$). Such large capital gains do not reflect fundamentals.
- ▶ Optimists: Real interest rates were falling globally, increasing the PV of rents.
- ▶ Also, mortgage market was changing: more people were able to borrow, and borrow a bigger proportion of the value of the house.
- ▶ https://www.youtube.com/watch?v=qo4ExWEA1_k

U.S Housing Price-to-Rent Ratio



Source: FRED: CSUS HPISA: Case-Shiller Home Price Index. CUSR0000SEHA, Rent of Primary Residence.

Mortgage delinquencies by type

